

PSY405 - Lesson 28

FACTOR ANALYTIC TRAIT THEORY

(Raymond Cattell)

Raymond Cattell wanted to answer one simple question:

| “What exactly is personality, and how can we measure it scientifically?”

Instead of guessing or using philosophy, he used **statistics** (numbers) and a method called **factor analysis** to **discover personality traits**.

👉 His idea:

- Personality is made of **traits**
 - Traits can be **discovered from behavior**
 - We should use **data + mathematics**, not opinions
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1 FACTOR ANALYSIS (Foundation of the Theory)

◆ What is Factor Analysis? (Plain English)

Factor analysis is a **statistical method** that:

- Takes **many behaviors**
- Finds **patterns**
- Groups them into **hidden factors**

👉 These hidden factors = **traits**

◆ Correlation (Key Term)

Correlation = when two things **change together**

 Example:

- Tall people often weigh more → **positive correlation**
- More stress, less sleep → **negative correlation**

 Correlation coefficient:

- **+1** = perfect positive
 - **-1** = perfect negative
 - **0** = no relationship
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◆ How Cattell Used Factor Analysis (Step by Step)

1. Measure many people

- Behavior
- Questionnaires
- Tests

2. Correlate everything

- Compare every score with every other score
- Creates a **correlation matrix** (big table of numbers)

3. Find clusters

- Behaviors that "go together"

4. Name the factor

- This factor = **trait**

 **Cattell believed traits are the building blocks of personality**

◆ Types of Data Cattell Used

Type	Name	Meaning
L-data	Life record	Real-life behavior
Q-data	Questionnaire	Self-reports

Type	Name	Meaning
T-data	Test data	Objective tests

 Example:

- L-data → attendance record
- Q-data → "I am shy"
- T-data → reaction time test

2 BIOGRAPHICAL SKETCH (Why It Matters)

Cattell:

- Born **1905 in England**
- Witnessed **World War I injuries** as a child
- Became interested in **human behavior and suffering**
- Worked in:
 - University of London
 - Clark University
 - Harvard
 - University of Illinois

 His life experience pushed him toward:

- **Scientific psychology**
- **Measurement**
- **Genetics + environment**

3 THE NATURE OF PERSONALITY

 **What is Personality According to Cattell?**

Personality is what allows us to predict what a person will do in a given situation.

📌 That's why he created the formula:

$$R = f(P, S)$$

Symbol	Meaning
R	Response (behavior)
P	Person (traits)
S	Situation

👉 Behavior = Person + Situation

4 STRUCTURE OF TRAITS (Very Important)

◆ What is a Trait?

Trait = a **mental structure** inferred from consistent behavior

📌 Example:

- Always honest → trait of **honesty**
- Always anxious → trait of **anxiety**

A UNIQUE vs COMMON TRAITS

Type	Meaning	Example
Common traits	Shared by many people	Intelligence
Unique traits	Special to one person	Artistic obsession

◆ R-Technique vs P-Technique

Technique	What it studies
R-Technique	Many people, many traits
P-Technique	One person over time

📌 P-technique is useful in **clinical psychology**

B NO DEFINITION SURFACE vs SOURCE TRAITS

◆ Surface Traits

- What we **see**
- Behaviors that appear together

📌 Example:

- Talkative
- Friendly
- Smiles a lot

→ surface trait: **sociability**

◆ Source Traits (Very Important)

- **Underlying causes**
- Found using **factor analysis**
- More **stable and powerful**

📌 Sociability may come from source trait **Extraversion**

👉 Surface traits = symptoms

👉 Source traits = causes

C NO DEFINITION CONSTITUTIONAL vs ENVIRONMENTAL TRAITS

Type	Meaning
Constitutional	Genetic, biological

Type	Meaning
Environmental	Learned from culture

 Example:

- Nervous system sensitivity → constitutional
- Manners → environmental

D | TYPES OF TRAITS

1 Ability Traits

- How well you perform tasks
- **Intelligence**

Two Types:

Type	Meaning
Fluid Intelligence	Problem-solving (genetic)
Crystallized Intelligence	Learned knowledge

2 Temperament Traits

- Emotional style
- Mood, speed, calmness

 Example:

- Calm vs hot-tempered

3 Dynamic Traits (Motivation)

Dynamic traits explain:

| WHY we act

5 IMPORTANT DYNAMIC TRAITS

◆ Three Types

1. **Ergs**
 2. **Sentiments**
 3. **Attitudes**
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A NO GROUP ERGS (Biological Drives)

Ergs = inborn drives

📌 Examples:

- Hunger
- Sex
- Curiosity
- Aggression

👉 Similar to **instincts**

B NO GROUP SENTIMENTS (Learned Drives)

- Learned emotional systems
- Organized around objects or ideas

📌 Examples:

- Love for family
 - Patriotism
 - Religious devotion
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◆ The Self (Most Important Sentiment)

- Central organizing system
- Almost all attitudes relate to **self**

👉 Self integrates personality

CATTALL ATTITUDES (Specific Responses)

Attitude = response toward a specific object or situation

📌 Example:

| "I strongly want to marry"

Cattell measured attitudes using:

- Memory tests
 - Physiological reactions
 - Errors & forgetting
-

◆ Dynamic Lattice (Simple Meaning)

A hierarchy:

- Ergs → Sentiments → Attitudes

👉 Biological → learned → specific actions

6 SOURCES OF PERSONALITY DATA (Revisited)

Data Type	What it gives
L-data	Real behavior
Q-data	Inner thoughts
T-data	Objective behavior

📌 Cattell trusted traits found in **all three**

7 DEVELOPMENT OF PERSONALITY

Personality develops through:

- **Genetics**
 - **Environment**
 - **Learning**
 - **Culture**
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8 HEREDITY-ENVIRONMENT ANALYSIS

Cattell believed:

- Traits are shaped by **both**
 - He tried to **measure their contribution**
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9 SEARCH METHODS

◆ 16 Personality Factors (16PF)

- Famous personality test
- Measures **16 source traits**

◆ Culture Fair Intelligence Tests

- Reduce cultural bias
 - Measure pure reasoning
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10 SUMMARY (One-Page Meaning)

- ✓ Personality = predictable behavior
- ✓ Traits = basic units
- ✓ Factor analysis finds traits
- ✓ Source traits are more important

- ✓ Motivation comes from ergs → sentiments → attitudes
 - ✓ Personality must be measured using **data**
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EVALUATION (Why Cattell Matters)

- ✓ Most scientific personality theory
 - ✓ Used by many psychologists
 - ✓ Influenced modern testing
 - ✗ Complex and hard to understand
 - ✗ Overly statistical
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1. What is factor analysis?

Answer:

Factor analysis is a **statistical method** used to identify patterns among a large number of variables. In psychology, it helps to find **underlying traits (factors)** that explain why certain behaviors tend to occur together. Cattell used it to discover the building blocks of personality.

Explanation:

- Imagine you measure **100 behaviors** in 1000 people.
- Some behaviors tend to **occur together** (cluster).
- Factor analysis **mathematically finds these clusters**, and each cluster represents a **factor**, which Cattell called a **trait**.
- It is **data-driven**, not just theory or opinion.

Example:

- People who are talkative also tend to smile a lot and make friends easily → cluster = **Extraversion**

Tip:

Remember: **Factor analysis** = “finding hidden patterns in many behaviors.”
Think of it like organizing a messy wardrobe by color groups.

2. What is a trait according to Cattell?

Answer:

A trait is a **mental structure inferred from observed behavior**, which accounts for consistency in how a person acts. It is the basic **building block of personality**.

Explanation:

- Traits are **not directly visible**; we infer them by observing patterns in behavior over time.
- Traits can be **unique** (specific to an individual) or **common** (shared by many people).

Example:

- Honesty: If someone consistently tells the truth, we infer the trait **honesty**.
- Creativity: A person who paints, writes, and invents regularly has the trait **creativity**.

Tip:

Think: **Trait** = “**pattern behind behavior**.” Behavior = surface, trait = source.

3. What are unique and common traits?

Answer:

- **Unique traits** are characteristics specific to a single individual.
- **Common traits** are characteristics shared by many people in a culture or community.

Explanation:

- Unique traits make a person **special**.
- Common traits show **general human patterns**.

Example:

- Unique trait: Prefers to collect antique teapots.
- Common trait: Intelligence, sociability.

Tip:

Memory hook: "**Unique = me, Common = we.**"

4. What are surface traits and source traits?

Answer:

- **Surface traits** are clusters of visible behaviors that appear together.
- **Source traits** are the **underlying causes** of these behaviors, discovered through factor analysis. Source traits are more **stable and fundamental**.

Explanation:

- Surface traits = what people **see**.
- Source traits = what **causes** the visible behavior.

Example:

- Surface trait: Friendly, talkative, outgoing → sociability
- Source trait: Extraversion → explains why these behaviors occur together

Tip:

Think: **Surface = symptom, Source = cause.**

5. What are constitutional and environmental-mold traits?

Answer:

- **Constitutional traits** are inherited or **biologically based**.
- **Environmental-mold traits** are shaped by **culture, education, and experiences**.

Explanation:

- Cattell recognized that both **nature and nurture** influence personality.
- Some traits come from **genes**; some are learned.

Example:

- Constitutional: Nervous system sensitivity
- Environmental: Politeness, study habits

Tip:

Memory hook: "**Constitution = built-in, Environment = shaped.**"

6. What are ability, temperament, and dynamic traits?

Answer:

- **Ability traits:** Determine how well a person performs tasks (e.g., intelligence).
- **Temperament traits:** Emotional style and response patterns (e.g., calm or irritable).
- **Dynamic traits:** Motivational traits that **drive behavior** toward goals.

Explanation:

- Ability → skill

- Temperament → emotional nature
- Dynamic → motivation

Example:

- Ability: Good at math
- Temperament: Patient or anxious
- Dynamic: Desire to become a teacher (motivates study, practice, and teaching)

Tip:

Remember **ATD** → **A**bility, **T**emperament, **D**ynamic.

7. What are ergs, sentiments, and attitudes?

Answer:

These are **types of dynamic traits**:

- **Ergs**: Innate biological drives (e.g., hunger, sex).
- **Sentiments**: Learned drives or emotional systems (e.g., patriotism, love for family).
- **Attitudes**: Specific responses toward particular objects or situations.

Explanation:

- Ergs → core instincts
- Sentiments → organized, learned patterns of motivation
- Attitudes → concrete, situational behaviors influenced by ergs and sentiments

Example:

- Erg: Hunger → motivates eating
- Sentiment: Love for pets → learned emotional system

- Attitude: "I want to adopt a puppy now" → situational response

Tip:

Memory trick: **Ergs** → **Sentiments** → **Attitudes** = **core** → **learned** → **specific**.

8. What is the dynamic lattice?

Answer:

The **dynamic lattice** is Cattell's model showing the **hierarchy of motivational traits**: ergs → sentiments → attitudes. It explains how **basic drives translate into specific behaviors**.

Explanation:

- Motivations are structured.
- Core biological needs influence learned sentiments, which influence specific attitudes or actions.

Example:

- Erg: Hunger → Sentiment: Value of healthy eating → Attitude: Chooses salad over chips

Tip:

Think of it as a **ladder from instincts to specific choices**.

9. What are the major sources of data about personality in Cattell's theory?

Answer:

1. **L-data (Life Records)**: Actual behavior observed in life, school, or work.
2. **Q-data (Questionnaire)**: Self-reports of behaviors, thoughts, and feelings.

3. **T-data (Test data):** Behavior in controlled situations or objective tests.

Explanation:

- Using all three allows **cross-validation**.
- Traits are stronger if they appear across L, Q, and T data.

Example:

- L-data: Teacher rates student's friendliness
- Q-data: Student reports being friendly
- T-data: Observed in a social interaction test

Tip:

Memory: **L = Life, Q = Questionnaire, T = Test**

10. How did Cattell define personality?

Answer:

Personality is **what allows us to predict a person's behavior** in a given situation.

Mathematically:

$$R = f(P, S)$$

- R = Response (behavior)
- P = Person (traits)
- S = Situation

Explanation:

- Behavior = a combination of personality and context
- Personality alone is **not enough**; situation matters too

Example:

- Calm person in a friendly setting → behaves calmly

- Same person in an emergency → may react differently

Tip:

Think: **Behavior = Traits + Situation.**

Formula helps you remember the core idea.

11. What are the 16 Personality Factors (16PF)?

Answer:

The **16PF** is a personality test developed by Cattell to measure **16 source traits**, considered the building blocks of personality.

Explanation:

- Derived from **factor analysis of multiple behaviors**
- Useful in career guidance, clinical, and research settings

Example:

- Traits measured include: Warmth, Reasoning, Emotional Stability, Dominance, Liveliness, etc.

Tip:

Think **16PF = the practical application of Cattell's theory.**

12. What is the difference between R-technique and P-technique?

Answer:

- **R-technique:** Correlation among traits **across many individuals**
- **P-technique:** Correlation among traits **within one individual over time**

Explanation:

- R-technique → finds **general factors for a population**
- P-technique → studies **how traits interact in a single person**

Example:

- R-technique: Studying 1000 students to find common personality traits
- P-technique: Tracking one student's mood, attention, and sociability over a year

Tip:

R = "Research many people"

P = "Personal longitudinal study"

1. True / False MCQs

Q1. Factor analysis is a statistical method used to identify underlying traits in personality.

- **Answer:** True 
- **Explanation:** Cattell used factor analysis to find clusters of behaviors (traits) that are not directly observable.

Q2. Surface traits are more stable and fundamental than source traits.

- **Answer:** False 
- **Explanation:** Surface traits are visible behaviors and less stable; source traits are underlying, stable traits discovered via factor analysis.

Q3. Constitutional traits are shaped primarily by environmental experiences.

- **Answer:** False 
- **Explanation:** Constitutional traits are biologically determined; environmental-mold traits come from experiences/culture.

Q4. Ergs, sentiments, and attitudes are types of dynamic traits.

- **Answer:** True 

- **Explanation:** These traits are part of Cattell's motivational structure (dynamic lattice).

Q5. R-technique involves studying trait correlations within a single individual over time.

- **Answer:** False 

- **Explanation:** That's P-technique; R-technique studies correlations across multiple individuals.
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2. Conceptual / Theory-Based MCQs

Q6. According to Cattell, personality is:

- A) The sum of observable behaviors
- B) A predictable pattern of behavior in a given situation
- C) Only inherited traits
- D) A person's conscious thoughts

- **Answer:** B 

- **Explanation:** Personality = what allows prediction of behavior ($R = f(P, S)$).

Q7. Which of the following is an example of a **source trait**?

- A) Talkative, cheerful, sociable
- B) Extraversion
- C) Smiling frequently
- D) Wearing bright colors

- **Answer:** B 

- **Explanation:** Source traits are the underlying causes of observed behaviors. A, C, D are surface traits.

Q8. Which type of data is collected through self-report questionnaires?

- A) L-data

- B) Q-data
- C) T-data
- D) R-data

- **Answer:** B 

- **Explanation:** Q-data = self-ratings; L-data = life records; T-data = objective tests.

Q9. Which of the following is **not** a dynamic trait?

- A) Ergs
- B) Attitudes
- C) Temperament
- D) Sentiments

- **Answer:** C 

- **Explanation:** Temperament is emotional style (constitutional), not motivational.

Q10. Cattell's 16PF test primarily measures:

- A) Surface traits
- B) Source traits
- C) Attitudes
- D) Life records

- **Answer:** B 

- **Explanation:** 16PF measures 16 source traits, the core building blocks of personality.

3. Situational / Applied MCQs

Q11. A teacher observes that a student is consistently cheerful, talkative, and helpful in class. Using Cattell's theory, this cluster of behaviors represents:

- A) Source traits
- B) Surface traits

C) Ergs

D) Sentiments

- **Answer:** B 

- **Explanation:** Observable patterns are surface traits; underlying cause might be **Extraversion** (source trait).

Q12. A person has a strong motivation to succeed in their career. According to Cattell, this motivational pattern could be explained by:

A) Attitudes only

B) Ergs, sentiments, and attitudes

C) Surface traits

D) Constitutional traits only

- **Answer:** B 

- **Explanation:** Dynamic traits (ergs, sentiments, attitudes) drive motivation and goal-directed behavior.

Q13. A psychologist collects school records (attendance, grades), self-reports, and objective tests to study personality. Which statement is correct?

A) L-data, Q-data, T-data are all used to find source traits

B) Only L-data is reliable

C) Objective tests are irrelevant in factor analysis

D) Q-data cannot show unconscious traits

- **Answer:** A 

- **Explanation:** Cattell combined L, Q, T data to validate source traits across methods.

Q14. A student is measured multiple times over a year to see how their anxiety and sociability change. Which technique is being used?

A) R-technique

B) P-technique

C) 16PF test

D) Dynamic lattice analysis

- **Answer:** B 

- **Explanation:** P-technique tracks traits **within a single person** over time.

Q15. If a person's sociability is influenced by both genetic temperament and cultural learning, which types of traits are involved?

A) Surface + source

B) Ergs + sentiments

C) Constitutional + environmental-mold

D) Ability + dynamic

- **Answer:** C 

- **Explanation:** Constitutional = genetic, Environmental-mold = shaped by culture and experience.

4. Tricky / Exam-Oriented MCQs

Q16. Which of the following is **true** about surface and source traits?

A) Surface traits are more fundamental than source traits

B) Source traits are visible behaviors

C) Surface traits are clusters of behaviors; source traits are underlying factors

D) Source traits cannot be measured

- **Answer:** C 

- **Explanation:** Surface = observable behaviors; Source = underlying causes revealed through factor analysis.

Q17. Fluid intelligence differs from crystallized intelligence in that:

A) Fluid intelligence is learned; crystallized is innate

B) Fluid intelligence is inherited; crystallized is learned

C) Both are innate

D) Both are learned

- **Answer:** B 

- **Explanation:** Fluid = problem-solving ability (genetic), Crystallized = knowledge gained from learning/experience.

Q18. If a researcher finds the same factor structure in L, Q, and T data, this indicates:

- A) Measurement error
- B) The factor is likely a true source trait
- C) Only surface traits are valid
- D) Attitudes are irrelevant

- **Answer:** B 

- **Explanation:** Consistency across data sources validates **true source traits**.

Q19. A person has a strong biological drive for food, develops a sentiment valuing healthy eating, and chooses salad for lunch. This sequence illustrates:

- A) Attitudes → Sentiments → Ergs
- B) Ergs → Sentiments → Attitudes
- C) Surface traits → Source traits → Dynamic traits
- D) Constitutional → Environmental → Dynamic

- **Answer:** B 

- **Explanation:** Dynamic lattice hierarchy: Ergs (instinct) → Sentiments (learned values) → Attitudes (specific action).

Q20. Which statement reflects Cattell's view of personality?

- A) Personality is only inherited and cannot be measured
- B) Personality is a collection of attitudes only
- C) Personality allows prediction of behavior in given situations
- D) Personality is fixed and unchangeable

- **Answer:** C 

- **Explanation:** Cattell emphasized **predictive utility**: behavior = f(person, situation).

Cheat Sheet

1 Factor Analysis

- **Definition:** Statistical method to identify **hidden patterns (factors/traits)** in observed behaviors.
- **Keyword:** Cluster analysis, correlation
- **Example:** Talkative + cheerful + friendly → cluster = **sociability**
- **Memory Hook:** "Factor = pattern behind behavior"
- **Mnemonic:** "FAP" → Factor Analysis Predicts

Steps of Factor Analysis:

1. Measure many people on many behaviors
 2. Create correlation matrix
 3. Find clusters of correlated behaviors
 4. Label clusters = **source traits**
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2 Personality (Cattell)

- **Definition:** Personality is what allows prediction of **behavior in a given situation**.
 - **Formula:** $R = f(P, S)$ → Response = function of Person + Situation
 - **Example:** Calm person in friendly setting → behaves calmly; in emergency → may act differently
 - **Memory Hook:** "Behavior = Traits + Situation"
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3 Traits

Term	Definition	Example	Keywords
Trait	Mental structure inferred from behavior	Honesty, creativity	Pattern, building block

Term	Definition	Example	Keywords
Unique trait	Specific to individual	Collects antique teapots	"Me" trait
Common trait	Shared by many	Intelligence, sociability	"We" trait
Surface trait	Observable behaviors	Talkative, cheerful	Symptom, visible
Source trait	Underlying cause	Extraversion	Fundamental, hidden
Constitutional	Biologically based	Nervous system sensitivity	Nature
Environmental-mold	Shaped by culture/learning	Manners	Nurture

Memory Hook:

- **Surface = symptom, Source = cause**
- **Unique = Me, Common = We**

4 Types of Traits (Cattell)

Type	Meaning	Example
Ability	Skill or task performance	Intelligence, problem-solving
Temperament	Emotional style	Calm, anxious, cheerful
Dynamic	Motivation & goal-directed behavior	Desire to become a doctor

Mnemonic: ATD → Ability, Temperament, Dynamic

5 Dynamic Traits Hierarchy (Dynamic Lattice)

Level	Definition	Example
Ergs	Innate biological drives	Hunger, curiosity, sex
Sentiments	Learned emotional systems	Love for family, patriotism
Attitudes	Specific responses	"I want to adopt a puppy"

Memory Hook: "Ergs → Sentiments → Attitudes = Core → Learned → Specific"

6 Major Sources of Personality Data

Data Type	Meaning	Example
L-data	Life record, observed behavior	School grades, court records
Q-data	Questionnaire / self-report	"I am shy"
T-data	Objective tests	Reaction time test, puzzle solving

Memory Hook: L = Life, Q = Questionnaire, T = Test

7 R-Technique vs P-Technique

Technique	Meaning	Example
R-Technique	Correlations across many people	Study 1000 students to find common traits
P-Technique	Correlations within one person over time	Track one student's anxiety over a year

Memory Hook: R = Research population, P = Personal longitudinal

8 16 Personality Factors (16PF)

- **Definition:** Test to measure **16 source traits** (core building blocks)
- **Examples of source traits:** Warmth, Reasoning, Emotional Stability, Dominance, Liveliness
- **Use:** Career guidance, research, clinical psychology
- **Memory Hook: "16PF = Source trait test"**

9 Intelligence Types (Ability Traits)

Type	Meaning	Example
Fluid	Problem-solving, general reasoning (genetic)	Solve novel puzzles

Type	Meaning	Example
Crystallized	Knowledge accumulated from learning	School subjects, vocabulary

Memory Hook: Fluid = Flow/innate, Crystallized = Learned/frozen knowledge

10 Quick Mnemonics for Exam

- **Trait hierarchy: Surface = symptom, Source = cause**
- **Dynamic traits: Ergs → Sentiments → Attitudes = Core → Learned → Specific**
- **ATD → Ability, Temperament, Dynamic**
- **Data types: L = Life, Q = Questionnaire, T = Test**
- **Techniques: R = Research population, P = Personal longitudinal**
- **Personality formula: $R = f(P, S) \rightarrow \text{Behavior} = \text{Traits} + \text{Situation}$**

11 Keywords Summary

Keyword	Meaning
Factor	Trait discovered via factor analysis
Correlation	Two behaviors vary together
Life Record (L-data)	Real-world behavior
Questionnaire (Q-data)	Self-reports
Objective test (T-data)	Lab or controlled behavior
Dynamic lattice	Hierarchy of motivational traits
Ergs	Biological drives
Sentiments	Learned drives
Attitudes	Specific responses